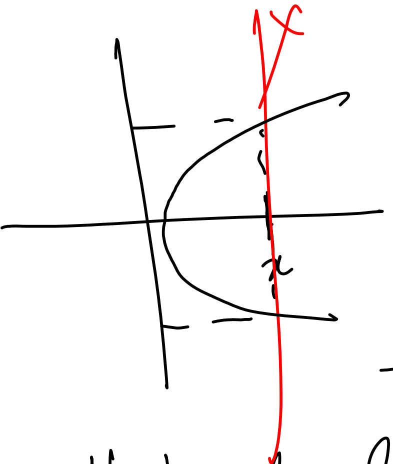
Unit 1.3 What is a function? p 10–23.

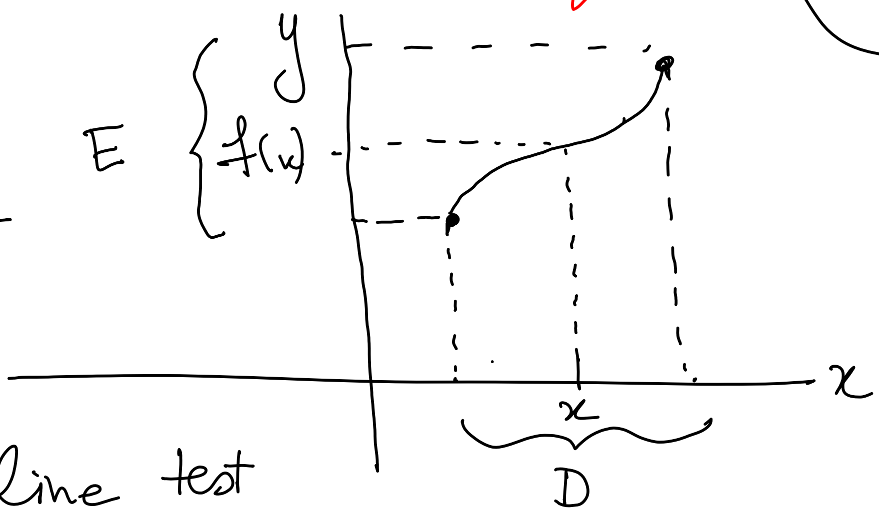
Definition:

A function f is a rule that assigns to each element x in a set D exactly one element, called $f(x)$, in a set E.

domain

range





• Vertical line test

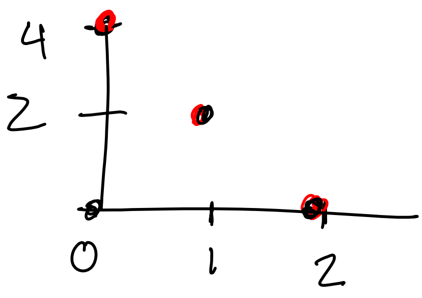
Four ways to represent a function:

• Verbally (in words)

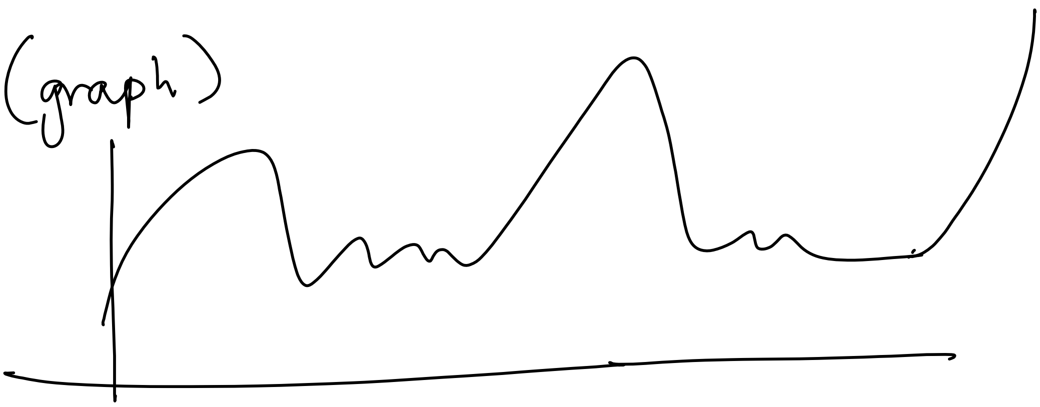
The more I see you the more I like you.

• numerically (table)

x	0	1	2
y	4	2	0



• visually (graph)



• algebraically (formula)

$$f(x) = x^2 - x + 2$$

Finding the domain and range of a function.

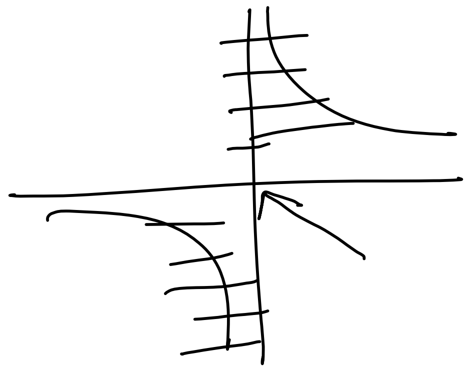
Domain: Use the formula

Range: Use the graph.

Eg: Find the domain and range of

1. $f(x) = \frac{1}{x}$

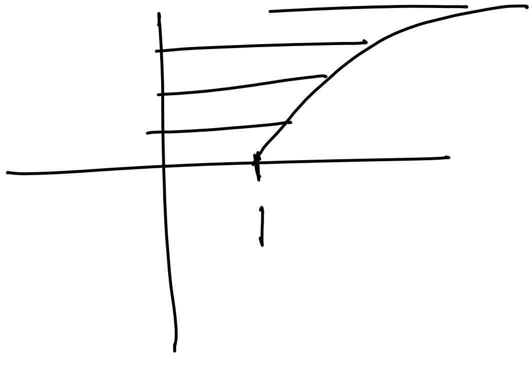
Domain: $\mathbb{R} \setminus \{0\}$



Range: $\mathbb{R} \setminus \{0\}$

2. $f(x) = \sqrt{(x-1)}_{+}$

Domain: $x - 1 \geq 0 \Rightarrow x \geq 1 \Rightarrow x \in [1, \infty)$



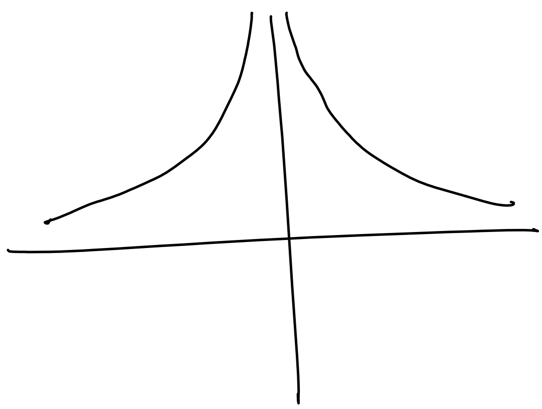
Range: $[0, \infty)$.

$\boxed{y \geq 0}$

3. $f(t) = \frac{1}{t^2}$

Domain: $\mathbb{R} \setminus \{0\}$

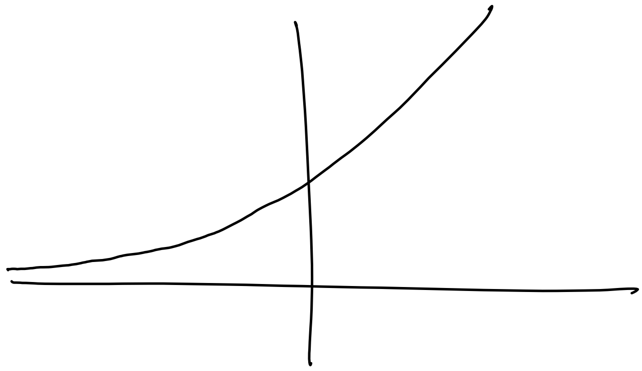
Range: $(0, \infty)$



4. $f(x) = 2^x$

Domain: $\mathbb{R}$.

Range: $(0, \infty)$



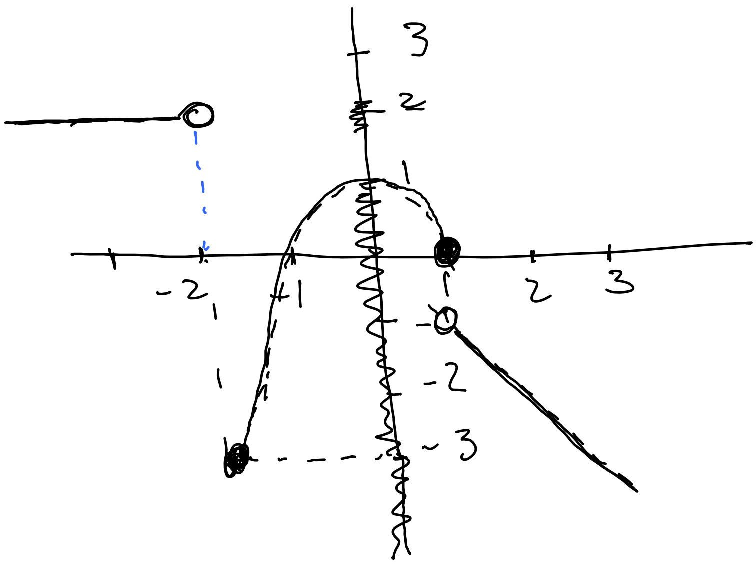
<u>Piece-wise</u> defined functions

Eg. $f(x) = \begin{cases} 2 & \text{if } x < -2 \\ -x^2+1 & \text{if } -2 \leq x \leq 1 \\ -x & \text{if } x > 1 \end{cases}$

<u>Domain:</u> $\mathbb{R}$.

<u>Range:</u>

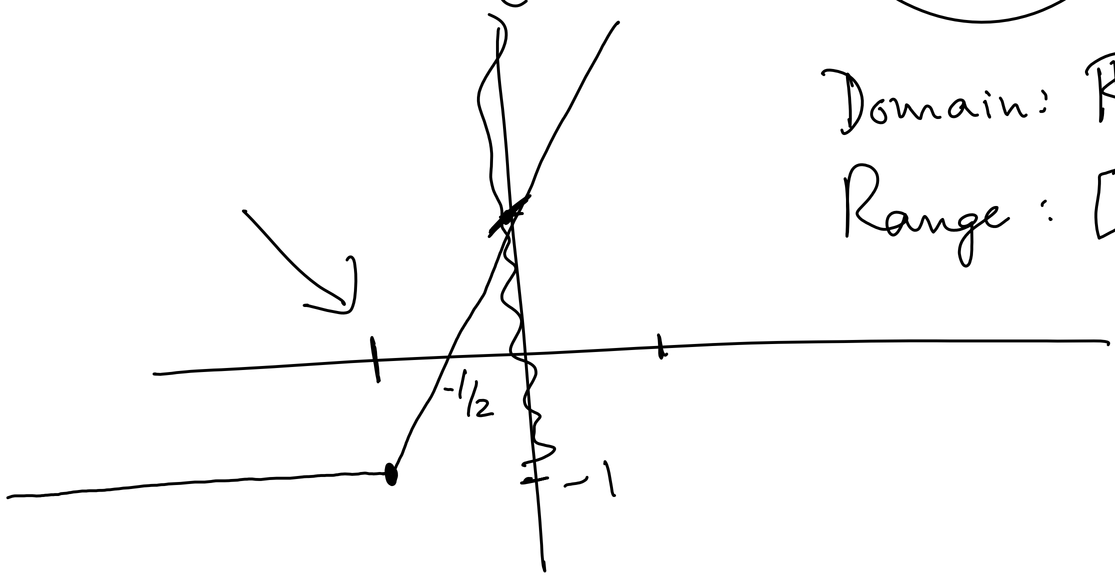
$(-\infty, 1] \cup \{2\}$



2. $f(x) = x + |x+1|$

$$= \begin{cases} x + x+1 & \text{if } x+1 \geq 0 \\ x - (x+1) & \text{if } x+1 < 0 \end{cases}$$

$$= \begin{cases} 2x+1 & \text{if } x \geq -1 \\ -1 & \text{if } x < -1. \end{cases}$$



Domain: $\mathbb{R}$

Range: $[-1, \infty)$.

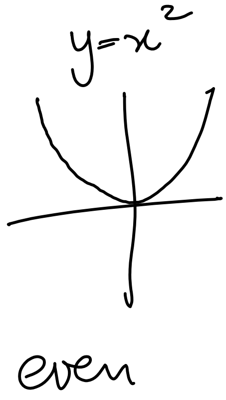
p 10 - 23.

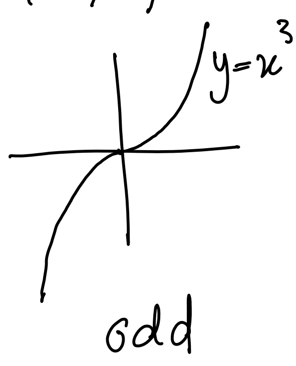
2, 4, 6, ...

1, 3, 5, ...

Symmetry of functions

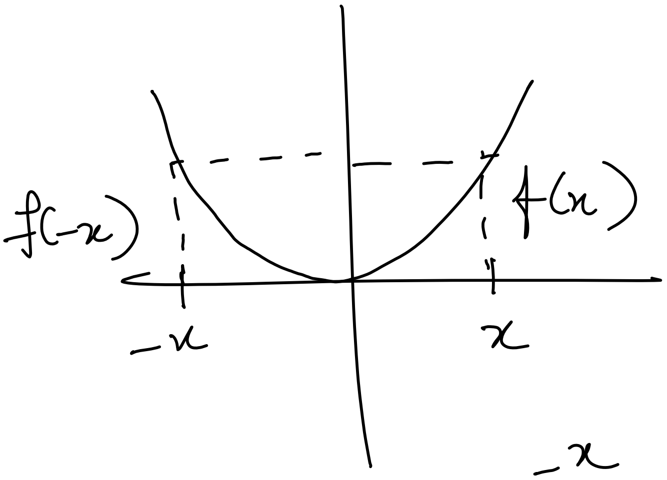
Even / Odd functions

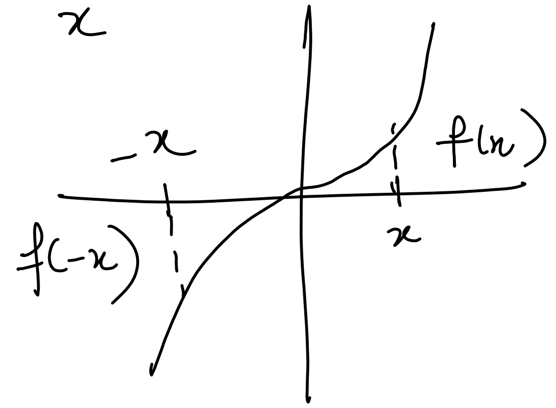




f is <u>even</u> if

$f(-x) = f(x)$
for all in the domain of f.





f is odd if $f(-x) = -f(x)$
for all x in the domain of f.

Eg. Is the function even, odd or neither

1. $f(x) = \dfrac{x^3}{x^2+1}$ Odd!

$f(-x) = \dfrac{(-x)^3}{(-x)^2+1} = \dfrac{-x^3}{x^2+1} = -f(x)$

2. $f(x) = x+3$

$f(-x) = -x+3$ Neither even nor odd.

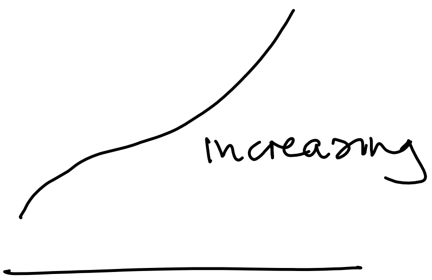
3. $f(x) = x^2+1$

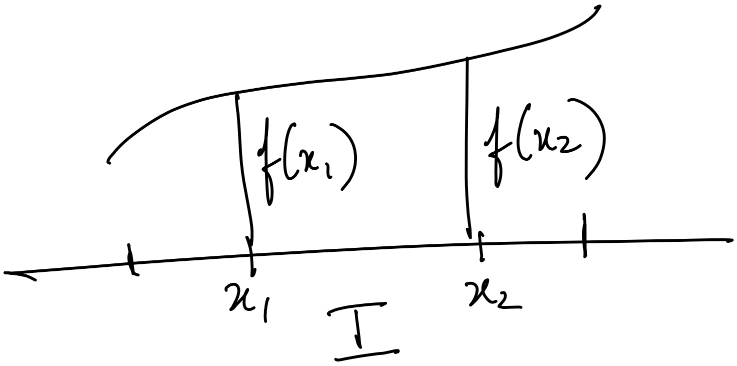
$f(-x) = (-x)^2+1 = x^2+1 = f(x)$ Even

4. $f(x) = |x|$

$f(-x) = |-x| = |x| = f(x)$ Even.

Increasing / Decreasing functions





$f(x_2) > f(x_1)$

<u>Def</u>: f is <u>increasing</u> on an interval I
if for any $x_1, x_2 \in I$ such that
$x_1 < x_2$ it holds that $f(x_2) > f(x_1)$

f is <u>decreasing</u> on an interval I
if for any $x_1, x_2 \in I$ such that
$x_1 < x_2$ it holds that $f(x_2) < f(x_1)$
$\cancel{x_1 > x_2}$ No!